ORL

ORL
DOI: 10.1159/000505130
Received: August 13, 2019
Accepted after revision: November 28, 2019
Published online: February 7, 2020
© 2020 S. Karger AG, Basel
www.karger.com/orl

Research Article

Tracheostomy Complications in the Emergency Department: A National Analysis of 38,271 Cases

Maxwell P. Kligerman[a] Anirudh Saraswathula[b] Rosh K. Sethi[c] Vasu Divi[a]

[a]Department of Otolaryngology – Head and Neck Surgery, Stanford University, Stanford, CA, USA; [b]Department of Otolaryngology, Johns Hopkins University, Baltimore, MD, USA; [c]Department of Otolaryngology, University of Michigan, Ann Arbor, MI, USA

Keywords
Tracheostomy · Complications · Emergency department · Head and neck cancer

Abstract
Background: Greater than 100,000 tracheotomies are performed annually in the USA, yet little is known regarding patients who present to the emergency department (ED) with tracheostomy complications. ***Objectives:*** To characterize patient and hospital characteristics, outcomes, and charges associated with tracheostomy complications and to identify predictors of admission and mortality. ***Methods:*** The 2009–2011 Nationwide Emergency Department Sample (NEDS) was queried for patients with a principle diagnosis of tracheostomy complication. A descriptive analysis was performed and multivariable logistic regression was used to identify predictors of admission and mortality. ***Results:*** A total of 69,371 nationwide visits to the ED had tracheostomy complication as an associated ICD-9 diagnosis, of which 55.2% (n = 38,293) carried a primary diagnosis of tracheostomy complication. Unspecified tracheostomy complications were most common (61.4%), followed by mechanical complications (31.3%), and lastly by tracheostomy infections (7.3%). Pediatric patients were significantly more likely to have tracheostomy infections than adults ($p < 0.0001$). A total of 35.5% of patients with tracheostomy complications were admitted to the hospital, and death occurred with 1.4% of visits. Patients from higher-income ZIP codes had increased odds of admission (adjusted odds ratio [OR]: 1.35; $p = 0.0009$), as did patients with tracheostomy infections (OR: 4.425; $p < 0.0001$). Patients with tracheostomy infections (OR: 3.14; $p = 0.0062$) and unspecified tracheostomy complications (OR: 2.00; $p = 0.0076$) had increased odds of mortality. ***Conclusion:*** These findings may help improve overall outcomes amongst patients with tracheostomies by preventing unnecessary ED admissions and improving healthcare provider preparedness and awareness.
© 2020 S. Karger AG, Basel

Maxwell P. Kligerman
Department of Otolaryngology – Head and Neck Surgery, Stanford University
801 Welch Road
Stanford, CA 94304 (USA)
Maxwell3@stanford.edu

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM

Introduction

Over 100,000 tracheotomies are performed annually in the USA [1]. A record number of patients with tracheostomies are now being discharged from the hospital, creating a particularly high-risk population for future complications [2]. It is not surprising, therefore, that tracheostomy-related complications are on the rise [3]. Emergency department (ED) healthcare providers are now frequently treating and triaging patients with tracheostomies who present with a variety of complications including infection, bleeding, plugging, dislodged tubes, and tracheal or stomal stenosis [4, 5].

Despite the increasing number of visits by patients with tracheostomies, these visits and patients alike remain poorly understood. A better characterization of the patient population presenting to EDs with tracheostomy complications would allow for improved resource allocation, prevention strategies, and provider education. This study utilizes a nationally drawn sample of ED visits in the USA to investigate patient and hospital characteristics, outcomes, and charges amongst those patients presenting with tracheostomy complications.

Materials and Methods

The 2009–2011 Nationwide Emergency Department Sample (NEDS), published by the Healthcare Cost and Utilization Project (HCUP) and sponsored by the Agency for Healthcare Research and Quality (AHRQ), was queried for patients with tracheostomy complications who presented to the ED.

The NEDS is a publicly available, all-payer ED database in the USA. For the years between 2009 and 2011, it contains discharge abstracts for 86,866,759 ED patient visits at 951 hospitals in 29 states and provides a 20% stratified sample of all hospital-based EDs in the country. Individual ED encounters can be weighted according to the sample design. Therefore, from 2009 to 2011, the NEDS provides a national weighted sample for over 388 million ED patient visits, representative of all ED visits across the country.

ICD-9 diagnosis codes were queried for tracheostomy complications. "Tracheostomy site infection" (ICD-9-CM code 519.01), "mechanical complication including tracheal stenosis" (code 519.02), and "other complications including tracheostomy hemorrhage and tracheo-esophageal fistula" (codes 519.00 and 519.09) were specifically included [1]. "Infection" refers specifically to soft tissue infection around the stoma and does not include tracheitis. The NEDS database was queried initially for visits with an ICD-9 diagnosis of tracheostomy complication in any ICD-9 diagnostic position. This cohort was then further analyzed for visits with an ICD-9 diagnosis of tracheostomy complication in only the first diagnostic position. A descriptive analysis was then performed on this second cohort of patients with a primary diagnosis of tracheostomy complication to characterize patient age, age category, gender, national quartile of median household income from the patient's ZIP code (as defined by the HCUP), insurance payer, urban-rural location of the patient's residence, day of the week of visit, total charge for ED visit, hospital teaching status, hospital geographic region, hospital admission, and mortality.

The patient demographics, visit characteristics, and hospital characteristics were compared. Patients with an insurance payer status of "self-pay" or "no charge" were classified as uninsured. Significant predictors of admission and mortality were determined in separate multivariable logistic regression models using median ZIP code income, insurance type, hospital location, patient age, gender, hospital teaching status, day of the week, and type of complication as a priori covariates in the regression model. Differences in proportions were assessed using Wald χ^2 tests. Data manipulation and analysis were performed using SAS v9.4 (SAS Institute, Cary, NC, USA).

Results

Tracheotomy Complications in the ED

Between 2009 and 2011, a nationwide total of 69,371 visits to the ED had an associated ICD-9 diagnosis related to tracheostomy complications. Of these cases, 55.2% (n = 38,293)

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM

Table 1. Demographic data

	Infection	Mechanical complication	Unspecified	Total	*p* value
Age category					<0.0001
0–17 years	1,064 (37.9%)	573 (4.7%)	1,828 (7.8%)	3,466 (9.1%)	
18–44 years	359 (12.8%)	1,856 (15.5%)	3,275 (13.9%)	5,490 (14.3%)	
45–64 years	836 (29.7%)	4,907 (50.0%)	9,691 (41.3%)	15,434 (40.3%)	
≥65 years	548 (19.5%)	4,636 (38.7%)	8,697 (37.0%)	13,881 (36.3%)	
Gender					0.6745
Male	1,671 (59.5%)	6,844 (57.2%)	13,529 (57.6%)	22,044 (57.7%)	
Female	1,137 (40.5%)	5,118 (42.8%)	9,938 (42.4%)	16,193 (42.3%)	
Median income ZIP code					
USD 1–38,999	934 (34.2%)	4,032 (34.5%)	8,060 (35.1%)	13,027 (34.8%)	0.0908
USD 39,000–47,999	769 (28.1%)	695 (27.7%)	6,606 (28.8%)	10,612 (28.4%)	
USD 48,000–63,999	594 (21.7%)	2,322 (19.9%)	4,932 (21.5%)	7,849 (21.0%)	
USD ≥64,000	435 (15.9%)	2,089 (17.9)	3,375 (14.7%)	5,899 (15.8%)	
Primary insurance payer					0.0002
Medicare (1)	855 (30.5%)	6,326 (53.0%)	11,398 (48.6%)	18,580 (48.7%)	
Medicaid (2)	1,323 (47.2%)	3,014 (25.3%)	6,439 (27.4%)	10,776 (28.2%)	
Private (3)	419 (15.0%)	1,838 (15.4%)	4,063 (17.3%)	6,321 (16.6%)	
Self-pay/no charge (4)	118 (4.2%)	440 (3.7%)	848 (3.6%)	1,407 (3.7%)	
Other (5)	86 (3.1%)	316 (2.6%)	689 (2.9%)	1,089 (2.9%)	
Patient's residence					0.0039
Large central, metropolitan	1,076 (38.6%)	4,156 (35.0%)	6,884 (29.5%)	12,117 (31.9%)	
Large fringe, metropolitan	624 (22.4%)	2,425 (20.4%)	4,595 (19.7%)	7,645 (20.1%)	
Medium metropolitan	393 (14.1%)	2,191 (18.4%)	5,016 (21.5%)	7,601 (20.0%)	
Small metropolitan	209 (7.5%)	1,095 (9.2%)	2,177 (9.3%)	3,481 (9.1%)	
Micropolitan	329 (11.8%)	1,259 (10.6%)	2,897 (12.4%)	4,485 (11.8%)	
Neither micropolitan nor metropolitan	156 (5.6%)	761 (6.4%)	1,751 (7.5%)	2,669 (7.0%)	
Hospital location					0.0039
Northeast (1)	432 (15.4%)	2,392 (20.0%)	3,958 (16.8%)	6,782 (17.7%)	
Midwest (2)	482 (17.2%)	3,127 (26.1%)	6,634 (28.2%)	10,243 (26.7%)	
South (3)	1,404 (50.0%)	4,706 (39.3%)	9,540 (40.6%)	6,782 (40.9%)	
West (4)	489 (17.4%)	1,747 (14.6%)	3,359 (14.3%)	5,595 (14.6%)	

carried a primary ICD-9 diagnosis of tracheostomy complication. This constitutes an annual average of 12,757 ED visits for primary tracheostomy complications. Among the patients with a primary tracheostomy complication, the mean age was 54.7 years (standard error [SE] = 0.56 years) and the majority was male (57.7%). Pediatric patients accounted for 9.05% (n = 3,466) of the ED visits. Additional demographic information described by type of complication is available in Table 1.

Complication Type

Unspecified tracheostomy complications were most common overall (61.4%; n = 23,491), followed by mechanical complications (31.3%; n = 11,972), and lastly by tracheostomy infections (7.3%; n = 2,807). Amongst the pediatric patients, unspecified complications were most common (52.7%; n = 1,828), followed by infection (30.7%; n = 1,064) (Fig. 1).

Hospital Admission

A total of 35.5% (n = 13,544) of the patients with primary tracheostomy complications were admitted to the hospital. Patients from higher-income ZIP codes (OR: 1.35; p = 0.0009),

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM

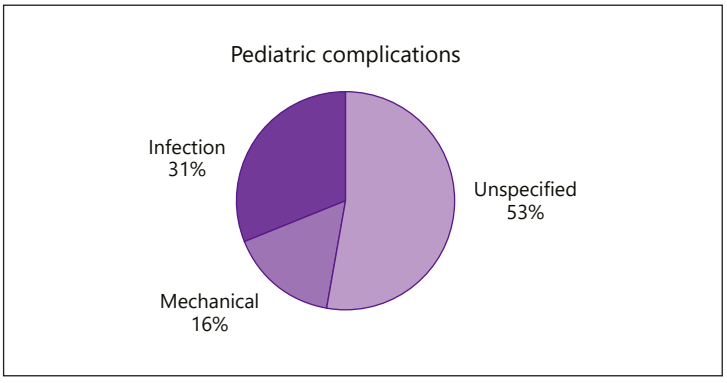

Fig. 1. The most common forms of tracheostomy complications amongst pediatric patients.

with tracheostomy infections (OR: 4.425; $p < 0.0001$), and with Medicaid insurance (OR: 2.79; $p < 0.001$) had increased odds of hospital admission (Table 2).

Hospital Mortality

A total of 1.4% (n = 521) of the visits associated with primary tracheostomy complications resulted in death. The vast majority of deaths were amongst adult patients (n = 508; 96.6%) as compared to pediatric patients (n = 13; 3.4%). Patients with tracheostomy infections (OR: 3.14; $p = 0.0062$) and with unspecified tracheostomy complications (OR: 2.00; $p = 0.0076$) had increased odds of mortality as compared to patients with mechanical complications (Table 2). Insurance and the median ZIP code income quartile were not significant predictors of mortality.

Associated Head and Neck Cancer Diagnoses

An associated head and neck cancer diagnosis in any ICD-9 code position occurred with 7.8% (n = 2,999) of the primary tracheostomy-related ED visits. The most common cancer subsites were the larynx (49.5%; n = 1,485), hypopharynx (28.5%; n = 854), and oral cavity (12.0%; n = 361). Patients with head and neck cancer were more likely to have unspecified tracheostomy complications (71.3%) than those without head and neck cancer (60.5%) ($p > 0.0001$) and faced a marginally significantly higher overall mortality (2.5%) than those without head and neck cancer (1.28%) ($p = 0.0509$).

Insurance Status and Hospital Charges

The total charges between 2009 and 2011 for all primary tracheostomy-related complications were USD 64.7 million. The median charge per ED visit was USD 1,233.00 (SE USD 28.41) and the mean charge was USD 1,988.88 (SE USD 57.01). Medicare insurance was most common (48.7%), followed by Medicaid (28.2%), and then private insurance (16.6%). Uninsured patients constituted 3.7% of the visits, and other insurance plans that were neither Medicare nor Medicaid nor private constituted 2.9% of the visits. ED visits associated with tracheostomy infections ($p < 0.0001$) and Medicaid insurance ($p = 0.0008$) were the most expensive.

Discussion and Conclusion

From 2009 to 2011, 38,293 ED visits had a primary diagnosis of a tracheostomy complication. Unspecified tracheostomy complications were most common, followed by mechanical complications, and lastly by tracheostomy infections. While the ICD-9 diagnostic codes do not provide much detail regarding the types of condition associated with each diagnostic group,

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM

Table 2. Predictors of admission and mortality

Variables	Hospital admission			Mortality		
	OR	95% CI	*p* value	OR	95% CI	*p* value
Complication type						
Mechanical Complication (2)	Ref.	–	–	Ref.	–	–
Infection (1)	4.425	3.315–5.906	<0.0001	3.139	1.384–7.122	0.0062
Unspecified (3)	1.085	0.953–1.235	0.2159	2.001	1.203–3.329	0.0076
Median income ZIP code						
USD 1–38,999	Ref.	–	–	Ref.	–	–
USD 39,000–47,999	1.141	0.989–1.315	0.0699	1.004	0.581–1.735	0.9875
USD 48,000–63,999	1.297	1.102–1.528	0.0018	1.033	0.597–1.787	0.9075
USD ≥64,000	1.356	1.133–1.622	0.0009	1.216	0.695–2.126	0.4934
Insurance coverage						
Medicaid	Ref.	–	–	Ref.	–	–
Medicare	1.052	0.902–1.227	0.5212	1.555	0.878–2.754	0.1296
Private	0.853	0.715–1.016	0.0753	1.655	0.875–3.129	0.1211
Uninsured	0.371	0.261–0.527	<0.0001	0.244	0.032–1.852	0.1722
Other	0.966	0.671–1.390	0.8512	1.213	0.367–4.012	0.7511
Hospital region						
Northeast	Ref.	–	–	Ref.	–	–
Midwest	0.743	0.595–0.928	0.0089	0.408	0.185–0.901	0.0266
South	0.826	0.676–1.010	0.0621	1.177	0.712–1.946	0.5254
West	0.876	0.701–1.094	0.2427	1.594	0.913–2.784	0.1013
Age						
Per 1 additional year	0.998	0.995–1.001	0.2551	1.013	1.002–1.024	0.0176
Gender						
Female	Ref.	–	–	Ref.	–	–
Male	0.978	0.880–1.087	0.6757	1.081	0.731–1.598	0.6964
Day of week						
Weekend	Ref.	–	–	Ref.	–	–
Weekday	1.091	0.979–1.217	0.1163	1.084	0.688–1.707	0.7272
Hospital teaching status						
Teaching, metropolitan	Ref.	–	–	Ref.	–	–
Nonteaching, metropolitan	0.777	0.672–0.899	0.0007	0.854	0.559–1.303	0.4634
Nonmetropolitan	0.497	0.409–0.604	<0.0001	0.706	0.345–1.446	0.3412

it is likely that unspecified tracheostomy complications are composed in large part of tracheostomy bleed events, while mechanical complications are composed in large part of tracheal or stomal stenosis, plugging events, or other tracheostomy tube malfunctions [1]. Further characterization of these groups may help improve resource allocation, identify high-risk groups, and demonstrate areas for education for healthcare providers.

Overall, mortality associated with tracheostomy complications was relatively low (1.4%; $n = 521$). Tracheostomy infection was the most significant predictor (OR: 3.14; $p = 0.0062$) of mortality, whereas socioeconomic factors like insurance and median ZIP code income were not associated with mortality. Interestingly, despite the fact that pediatric patients were over 6 times more likely to present with a tracheal site infection than adults, pediatric patients did not face increased odds of death overall. This is because the vast majority of total deaths that occurred were amongst adult patients (96.6%) as compared to pediatric patients (3.4%). A clear explanation as to why patients with tracheostomy infections faced increased odds of

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM

death remains unclear. It is possible that overall those patients who develop tracheostomy infections are worse or more likely to be ventilator dependent.

Other analyses of postoperative tracheostomy complications in pediatric patients have found infection to be common, but few others have shown it to be such an important driver of ED visits [6–8]. While the exact reason why pediatric patients face such a high risk of infection is unclear, it is possible that tracheostomies in pediatric patients present a unique challenge for healthcare providers to clean. A recent study evaluating the effect of a standardized wound care parent education program, for instance, resulted in a nearly 50% decrease in the rate of pediatric tracheostomy infections [9]. Widespread dissemination of such protocols may significantly decrease rates of tracheostomy infection, pediatric ED admission, and even mortality.

A significant proportion of patients with tracheostomy complications were admitted to the hospital (35.5%). Patients from higher-income ZIP codes faced increased odds of admission, whereas uninsured patients had the lowest odds of admission. One possible explanation for these findings is that patients from higher-income ZIP codes are more likely to visit the ED for life-threatening emergencies [10]. Uninsured patients, on the other hand, may have less access to primary care and may be more likely to use the ED for lower-acuity needs. Efforts to reduce preventable ED visits may focus on implementing multidisciplinary tracheostomy care teams prior to initial hospital discharge. Such teams, often composed of otolaryngologists, speech language pathologists, respiratory therapists, and outreach nurses, have been shown to decrease time to decannulation, improve patient knowledge, and decrease the length of hospital admission [11–13]. Special attention should be paid to ensuring that those patients who are readmitted are reintroduced to both an inpatient and an outpatient multidisciplinary tracheostomy team. Additional strategies focused on improving primary care provider comfort and knowledge regarding tracheostomy care may prevent unnecessary admissions [14–16].

Patients with an associated head and neck cancer diagnosis constitute a minority of all patients with tracheostomies who present to the ED (7.8%), but they represent a unique, high-risk group. The increased rate of unspecified tracheostomy complications, for instance, may be secondary to an increased risk of spontaneous hemorrhage and tracheostomy bleeds amongst patients with head and neck cancer [17, 18]. It is also likely that inherent risk factors in patients with head and neck cancer could make management in the emergency setting more difficult and contribute to the marginally significantly higher overall mortality (p = 0.0509) found in this study. Tumor burden, for instance, can cause severe or recalcitrant bleeding as well as upper airway obstruction in case of tracheostomy tube malfunction [19]. Difficult airway response teams and specialty algorithms in case of major bleeding have been shown to mitigate the risk in both scenarios and may be helpful adjuncts in EDs treating large numbers of patients with either tracheostomies or head and neck cancer [20–23].

This study has several key limitations. First, the CPT (Current Procedural Terminology) code classifications and the large proportion of "unspecified" problems provide little granularity and detail into the exact type of complication that occurred. In addition, this study was unable to identify the initial date of tracheostomy, and thus was unable to determine whether these visits represented acute postoperative complications or long-term sequalae. As with all retrospective database studies, there may be unmeasured confounders such as prior ED visits, referral patterns, and the reason for tracheostomy.

Patients with tracheostomy complications frequently visited EDs, and nearly one-third of the patients were admitted. While the overall mortality rates were relatively low, this analysis identified that patients with tracheostomy infection faced the highest odds of mortality and that pediatric patients were nearly 6 times more likely than adults to develop infections. Future efforts may focus on healthcare provider education on routine tracheostomy care as well as preventive strategies targeting high-risk groups.

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM

Statement of Ethics

This study was exempt from Stanford IRB submission because it included only deidentified data and did not qualify as research on human subjects.

Disclosure Statement

The authors have no conflict of interest.

Funding Sources

No funding was provided for this research paper.

Author Contributions

M.P. Kligerman contributed to the conception and design of the project, analysis, and drafting and revising of the manuscript; A. Saraswathula and R.K. Sethi contributed to the analysis as well as drafting and revising of the manuscript; V. Divi contributed to the conception and design of the work as well as revising of the manuscript.

References

1 Shah RK, Lander L, Berry JG, Nussenbaum B, Merati A, Roberson DW. Tracheotomy outcomes and complications: a national perspective. Laryngoscope. 2012 Jan;122(1):25–9.
2 Mehta AB, Syeda SN, Bajpayee L, Cooke CR, Walkey AJ, Wiener RS. Trends in tracheostomy for mechanically ventilated patients in the United States, 1993–2012. Am J Respir Crit Care Med. 2015 Aug;192(4):446–54.
3 Das P, Zhu H, Shah RK, Roberson DW, Berry J, Skinner ML. Tracheotomy-related catastrophic events: results of a national survey. Laryngoscope. 2012 Jan;122(1):30–7.
4 Johnson RF, Saadeh C. Nationwide estimations of tracheal stenosis due to tracheostomies. Laryngoscope. 2019 Jul;129(7):1623–6.
5 Hackeling T, Triana R, Ma OJ, Shockley W. Emergency care of patients with tracheostomies: a 7-year review. Am J Emerg Med. 1998 Nov;16(7):681–5.
6 Dal'Astra AP, Quirino AV, Caixêta JA, Avelino MA. Tracheostomy in childhood: review of the literature on complications and mortality over the last three decades. Braz J Otorhinolaryngol. 2017 Mar-Apr;83(2):207–14.
7 Shay S, Shapiro NL, Bhattacharyya N. Revisits after pediatric tracheotomy: airway concerns result in returns. Int J Pediatr Otorhinolaryngol. 2018 Jan;104:5–9.
8 Mahida JB, Asti L, Boss EF, Shah RK, Deans KJ, Minneci PC, et al. Tracheostomy placement in children younger than 2 years: 30-day outcomes using the National Surgical Quality Improvement Program Pediatric. JAMA Otolaryngol Head Neck Surg. 2016 Mar;142(3):241–6.
9 Gaudreau PA, Greenlick H, Dong T, Levy M, Hackett A, Preciado D, et al. Preventing complications of pediatric tracheostomy through standardized wound care and parent education. JAMA Otolaryngol Head Neck Surg. 2016 Oct;142(10):966–71.
10 Kligerman MP, Sethi RK, Kozin ED, Gray ST, Shrime MG. Morbidity and mortality among patients with head and neck cancer in the emergency department: a national perspective. Head Neck. 2019 Apr;41(4):1007–15.
11 Cameron TS, McKinstry A, Burt SK, Howard ME, Bellomo R, Brown DJ, et al. Outcomes of patients with spinal cord injury before and after introduction of an interdisciplinary tracheostomy team. Crit Care Resusc. 2009 Mar;11(1):14–9.
12 Tobin AE, Santamaria JD. An intensivist-led tracheostomy review team is associated with shorter decannulation time and length of stay: a prospective cohort study. Crit Care. 2008;12(2):R48.
13 Arora A, Hettige R, Ifeacho S, Narula A. Driving standards in tracheostomy care: a preliminary communication of the St Mary's ENT-led multi disciplinary team approach. Clin Otolaryngol. 2008 Dec;33(6):596–9.
14 Yelverton JC, Nguyen JH, Wan W, Kenerson MC, Schuman TA. Effectiveness of a standardized education process for tracheostomy care. Laryngoscope. 2015 Feb;125(2):342–7.
15 Agarwal A, Marks N, Wessel V, Willis D, Bai S, Tang X, et al. Improving knowledge, technical skills, and confidence among pediatric health care providers in the management of chronic tracheostomy using a simulation model. Pediatr Pulmonol. 2016 Jul;51(7):696–704.

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM

16 Dorton LH, Lintzenich CR, Evans AK. Simulation model for tracheotomy education for primary health-care providers. Ann Otol Rhinol Laryngol. 2014 Jan;123(1):11–8.
17 Sesterhenn AM, Iwinska-Zelder J, Dalchow CV, Bien S, Werner JA. Acute haemorrhage in patients with advanced head and neck cancer: value of endovascular therapy as palliative treatment option. J Laryngol Otol. 2006 Feb; 120(2):117–24.
18 Zhao LB, Shi HB, Park S, Lee DG, Shim JH, Lee DH, et al. Acute bleeding in the head and neck: angiographic findings and endovascular management. AJNR Am J Neuroradiol. 2014 Feb;35(2):360–6.
19 Pan CT, Lee LA, Fang TJ, Li HY, Liao CT, Chen IH. NBI flexible laryngoscopy targeted tissue sampling in head and neck cancer patients with difficult airways. Eur Arch Otorhinolaryngol. 2013 Jan;270(1):263–9.
20 Mark L, Lester L, Cover R, Herzer K. A decade of difficult airway response team: lessons learned from a hospital-wide difficult airway response team program. Crit Care Clin. 2018 Apr;34(2):239–51.
21 Mark LJ, Herzer KR, Cover R, Pandian V, Bhatti NI, Berkow LC, et al. Difficult airway response team: a novel quality improvement program for managing hospital-wide airway emergencies. Anesth Analg. 2015 Jul; 121(1):127–39.
22 Citardi MJ, Chaloupka JC, Son YH, Ariyan S, Sasaki CT. Management of carotid artery rupture by monitored endovascular therapeutic occlusion (1988–1994). Laryngoscope. 1995 Oct;105(10):1086–92.
23 Tuchman JB, Concetta Lupa M, Jooste EH. Emergency airway management of tracheo-innominate fistulas in the pediatric population. Can J Anaesth. 2010 Sep;57(9):865–7.

Downloaded by:
Université de Paris
193.51.85.197 - 2/15/2020 7:41:28 PM